Athari â€” Day 1 Report: Setup & Content Model

Owner: Baraka Mitigoa **Project:** Athari (NGO Project & Impact Portal) **Date:** 16 October 2026 **Local site:** https://athari.ddev.site **Admin login (local only):** `admin` / `admin`

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1. Summary

Day 1 turned an empty folder into a working Drupal 11 site with a complete data model.

- The local toolchain (WSL2, Docker Desktop, DDEV) was installed and verified.
- A Drupal 11.4.10 site was created in `C:\NGO\Athari`, running in DDEV containers, with Drush for the command line.
- Git was initialised, with a proper `.gitignore`, and five logical commits were made.
- Seven contrib modules (plus Devel for development) were installed and enabled.
- Two taxonomies (**Sector**, **Region**) and four content types (**Project**, **Location**, **Partner**, **Report**) were created with all the fields the plan requires.
- Pathauto now produces clean URLs such as `/projects/clean-water-gulu`.
- All configuration is exported to `config/sync`, so the whole site structure lives in Git.

Nothing was left incomplete from the Day 1 checklist. Two things were deliberately deferred: roles/permissions and sample data (Day 2), and a custom theme (Day 3).

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2. Everything created or changed

2.1 Project scaffolding

Item	What it is	Why it was needed
`.ddev/config.yaml`	DDEV project definition: name `athari`, type `drupal11`, docroot `web`, PHP 8.3, MariaDB 11.8.	Tells DDEV how to build the web and database containers.
`composer.json`, `composer.lock`	Lists every PHP package the site needs (Drupal core, Drush, contrib modules) and pins exact versions.	Anyone can rebuild identical code with `composer install`.
`web/`	The web root. Drupal core lives in `web/core` (not committed; Composer restores it).	Docroot served by DDEV.

Item	What it is	Why it was needed
`web/sites/default/settings.php`	Main Drupal settings file. I added `$settings['config_sync_directory'] = '../config/sync'` and an include of an optional `settings.local.php`.	Moves the config export outside the web root, as the plan requires, and gives a git-ignored place for secrets (Daraja keys on Day 4).
`web/sites/default/settings.ddev.php`	Generated by DDEV: database credentials and hash salt for the local environment.	Lets Drupal connect to the DDEV database. It is git-ignored.
`config/sync/`	Directory of YAML files describing the site's configuration.	Drupal's configuration management target. 138 files at install, 213 now.
`.gitignore`	Excludes `vendor/`, Drupal core and contrib code, uploaded files, `settings.local.php`, scaffold files, `.env`, OS junk and `.claude/`.	Keeps the repo small and prevents secrets being committed.
`scripts/day1-content-model.php`	A Drush script that builds the vocabularies, terms, content types, fields, displays and Pathauto patterns.	Repeatable and reviewable creation of the model. The exported config is the source of truth.
`docs/reports/`	This report and its PDF.	Mandatory daily report.

2.2 Taxonomies

Vocabulary	Terms	Purpose
Sector (`sector`)	Health, Education, WASH, Livelihoods, Protection	The thematic area of a project. Drives the filter and the "by sector" chart on Day 3.
Region (`region`)	4 top-level regions (Central, Eastern, Northern, Western), each with 5 child districts (20 districts)	Geography. Hierarchical, so a project can be tagged to a whole region or to a specific district. Drives the "by region" chart.

Districts used: Central â€“ Kampala, Wakiso, Mukono, Masaka, Mpigi. Eastern â€“ Jinja, Mbale, Soroti, Tororo, Busia. Northern â€“ Gulu, Lira, Arua, Kitgum, Adjumani. Western â€“ Mbarara, Kabarole, Kasese, Kabale, Hoima.

2.3 Content types and fields

All four types have revisions enabled and "Authored by/on" display switched off. "Promoted to front page" and "Sticky" are hidden from the edit forms, because the homepage will be built with Views on Day 3.

Partner (`partner`) â€" an organisation involved in projects.

Field	Machine name	Type	Notes
Name	`title`	Text (core)	The partner's name.
Logo	`field_logo`	Image	png/jpg/svg/webp, stored in `partners/` .
Website	`field_website`	Link	External URL only.

Location (`location`) â€" a project site.

Field	Machine name	Type	Notes
Name	`title`	Text (core)	Site name.
District	`field_district`	Entity reference â†' Region term	Required.
Coordinates	`field_coordinates`	Geofield	Required. Entered as latitude/longitude. Leaflet will plot it on Day 3.

Project (`project`) â€" the central content type.

Field	Machine name	Type	Notes
Title	`title`	Text (core)	
Description	`body`	Long text with summary	
Budget (USD)	`field_budget`	Decimal (14,2)	Minimum 0.
Start date / End date	`field_start_date` , `field_end_date`	Date	Date only, no time.
Status	`field_status`	List (text)	Planned / Active / Completed. Required.
Sector	`field_sector`	Reference â†' Sector term	Required.
Region	`field_region`	Reference â†' Region term	Required. Region or district.
Locations	`field_locations`	Reference â†' Location nodes	Unlimited. Autocomplete.
Partners	`field_partners`	Reference â†' Partner nodes	Unlimited. Autocomplete.

I made Locations and Partners multi-value: a real project often has several sites and several partners, and a single value would have been a limitation later.

Report (`report`) â€" a field report on a project.

Field	Machine name	Type	Notes
Title	`title`	Text (core)	
Project	`field_project`	Reference â†' Project	Required. Links each report to its project, the basis of all impact totals.
Report date	`field_report_date`	Date	Required.
Beneficiaries reached	`field_beneficiaries`	Integer	Required, minimum 0. Summed on Day 3.
Funds spent (USD)	`field_funds_spent`	Decimal (14,2)	Required. Summed on Day 3.
Narrative	`field_narrative`	Long text	Required.
Attachments	`field_attachments`	File	Unlimited. pdf, doc(x), xls(x), csv, images.

Each field also has a form display (the widget used when editing) and a view display (how it is shown on the page). These are config items too.

2.4 Pathauto URL patterns

Pattern	Applies to	Result
`/projects/[node:title]`	Project	`/projects/clean-water-gulu`
`/locations/[node:title]`	Location	`/locations/gulu-town`
`/partners/[node:title]`	Partner	`/partners/example-org`
`/reports/[node:title]`	Report	`/reports/q1-field-report`
`/sector/[term:name]`	Sector terms	`/sector/health`
`/region/[term:name]`	Region terms	`/region/gulu`

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3. Modules installed

Module	Version	Purpose	Use in Athari
Admin Toolbar (+ Tools)	3.6.3	Dropdown admin menu.	Faster navigation for staff and for building.
Pathauto	1.15.0	Automatic URL aliases from patterns.	Clean, readable URLs for every content item.
Token	1.17.0	Placeholders such as `[node:title]` .	Required by Pathauto; used in file paths.
Geofield	10.3.4	Latitude/longitude field type.	Stores Location coordinates.
Leaflet	10.4.13	Interactive maps using Leaflet.js.	Project map on Day 3.

Module	Version	Purpose	Use in Athari
Charts (+ Chart.js)	5.2.4	Chart rendering in Views.	Impact dashboard charts on Day 3.
Address	2.0.4	Structured postal address field.	Available for partner/contact details (not yet used on a field).
Devel, Devel Generate	5.5.0	Developer tools and dummy-data generation.	Dev only. Not for production.
Drush	13.8.0	Drupal command-line tool.	Installs, config export, cache clears.

Pulled in automatically: CTools (required by Pathauto), commerceguys/addressing (Address), GeoPHP (Geofield). `datetime_range` and the core Toolbar were also enabled as dependencies.

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4. Code walkthrough

There is no custom module yet (`athari_donate` starts on Day 4). The only custom code is the build script.

`scripts/day1-content-model.php` runs with `ddev drush php:script`, which boots Drupal and runs the script with full access to its APIs.

- `athari_add_field()` â€" a helper that does four things for one field: 1. creates **field storage** (`FieldStorageConfig`): the column(s) in the database, type and cardinality; 2. creates a **field instance** (`FieldConfig`): the field attached to one content type, with label, required flag and settings; 3. adds the field to the **form display** (`EntityFormDisplay`) with a widget; 4. adds it to the **view display** (`EntityViewDisplay`) with a formatter. It checks whether each piece exists first, so the script can be re-run safely.
- `athari_ref()` â€" builds the settings for an entity reference field: the target entity type and which bundles (vocabularies or content types) may be referenced.
- Vocabularies and terms â€" `Vocabulary::create()` and `Term::create()` . Districts are created with the region as their `parent` , giving the hierarchy.
- Content types â€" `NodeType::create()` for the four types.
- Pathauto â€" `PathautoPattern::create()` with a selection condition (`entity_bundle:node`) so each pattern only applies to its own type.

Data flow: the script writes Drupal *config entities* to the database. `drush cex` then exports them to YAML in `config/sync` . On another machine, `drush cim` reads the YAML back, so the same structure appears without re-running the script.

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5. How to use / test it

1. Start the project if needed: `ddev start` (from `C:\NGO\Athari`).
2. Open https://athari.ddev.site and log in at https://athari.ddev.site/user/login with `admin` / `admin` . (Or run `ddev drush uli` for a one-time login link.)
3. **Taxonomies:** Structure â†' Taxonomy (`/admin/structure/taxonomy`). Open **Region** to see the regions and their child districts.

4. **Content types:** Structure â†' Content types (`/admin/structure/types`). Open **Project** â†' *Manage fields* to see all fields; *Manage form display* to see the edit-form layout.
5. **Try it:** Content â†' Add content (`/node/add`). - Add a **Partner**, then a **Location** (coordinates e.g. latitude 2.7746, longitude 32.2990 for Gulu). - Add a **Project** that references the location and partner. - Add a **Report** that references the project. - After saving, check the URL: it should look like `/projects/your-title` .
6. **Pathauto:** Configuration â†' Search and metadata â†' URL aliases â†' Patterns (`/admin/config/search/path/patterns`).
7. Bulk-generate aliases for existing content any time: `ddev drush pathauto:aliases-generate create all` .

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6. Commands run

Command	What it does
`wsl --install --web-download`	Installs WSL2 and Ubuntu (web download after the first attempt failed).
`winget install DDEVFoundation.DDEV`	Installs DDEV on Windows.
`mkcert -install`	Creates a local certificate authority so `https://*.ddev.site` is trusted.
`ddev config --project-type=drupal11 --docroot=web --project-name=athari --php-version=8.3`	Writes `.ddev/config.yaml` .
`ddev start`	Pulls images, builds and starts the web, database and router containers.
`composer create-project drupal/recommended-project:^11 /tmp/athari-build --no-install` (inside the container)	Downloads the Drupal 11 project template into a temporary folder. See Issues.
`composer install`	Installs Drupal core and its dependencies from the lock file.
`ddev composer require drush/drush`	Adds Drush.
`ddev drush site:install standard ...`	Installs Drupal with the Standard profile and creates the admin user.
`ddev drush cex -y`	Exports active configuration to `config/sync` .
`ddev composer require drupal/admin_toolbar drupal/pathauto ...`	Downloads the contrib modules.
`ddev drush en admin_toolbar pathauto token geofield leaflet charts charts_chartjs address devel devel_generate`	Enables them.
`ddev drush php:script scripts/day1-content-model.php`	Builds the content model.
`ddev drush pathauto:aliases-generate create all`	Creates URL aliases for existing content and terms.

Command	What it does
`ddev mutagen sync`	Forces DDEV's file sync between container and Windows.
`git init -b main`, `git add`, `git commit`	Version control.

7. Drupal concepts explained

- **Docroot and Composer:** Drupal's public files live in `web/`; everything else (`vendor/`, `config/`, `composer.json`) stays outside the web server's reach, which is more secure. Composer downloads and versions all code; you never download modules by hand.
- **Content type, node, field:** A *node* is one piece of content. A *content type* (bundle) defines what fields a node has. A *field* has two halves: *storage* (database definition, shared across types) and an *instance* (its use on one bundle).
- **Entity reference:** A field that points to another entity (a taxonomy term or node) by ID. Project â†' Sector is a term reference; Report â†' Project is a node reference. This is how Drupal models relationships, and what Views aggregates over on Day 3.
- **Taxonomy:** Vocabularies of terms for classification. Region is hierarchical (parent/child) to model region â†' district.
- **Form display vs view display:** The form display decides which widget edits a field; the view display decides how the field is shown to visitors. Both are configuration.
- **Configuration management:** Site structure (types, fields, views) is stored as config in the database and can be exported to YAML (`cex`) and imported (`cim`). *Content* (nodes, terms) is not config, so it is not in Git. Day 2 seeds content with a script instead.
- **Pathauto and tokens:** Pathauto builds URL aliases from patterns; tokens like `[node:title]` are replaced with real values.
- **Geofield:** Stores geometry (here a point). The Leaflet module reads it to draw maps.
- **DDEV and Mutagen:** DDEV runs Drupal in Docker containers. On Windows it uses Mutagen to sync files between Windows and the container for speed; the sync is asynchronous, so a few seconds may pass before changes appear on the other side.

8. Issues and fixes

Issue	Cause	Fix
Docker, WSL2 and DDEV were missing at the start.	Fresh machine.	Installed them following the steps given (WSL2 â†' Docker Desktop â†' DDEV).
`wsl --install` failed: "connection with the server was reset".	Network drop during download.	Retried with `--web-download`; completed.
`winget install DDEV.DDEV` said "No package found".	DDEV's winget ID changed.	`winget search ddev` showed `DDEVFoundation.DDEV`.

Issue	Cause	Fix
`mkcert -install` printed a Java `keytool` error for Android Studio.	mkcert tried to add its certificate to Android Studio's Java store, which needs admin rights.	Ignored: the Windows trust store (used by browsers) was updated successfully.
`ddev composer create` wants an empty folder, but the folder already held `CLAUDE.md` and DDEV files.	Composer's create-project refuses non-empty targets.	Created the project in a temp folder inside the container, copied `composer.json` and `composer.lock`, then ran `composer install`.
First `drush/drush` install did not run.	A PowerShell pipe to `tail` (a Linux command) errored.	Re-ran without the pipe.
Some `ddev drush` calls hung when run from PowerShell.	Drush waited on input from the console.	Used `ddev exec "drush â€¦ < /dev/null"` and killed the stuck process.
`git status` showed no config changes straight after `drush cex`.	Mutagen had not synced the new files to Windows yet.	Ran `ddev mutagen sync` before committing.
URL aliases were missing for taxonomy terms.	The terms were created before the patterns existed.	Ran `pathauto:aliases-generate create all`.
Git warns "LF will be replaced by CRLF".	Windows line-ending setting.	Harmless; not changed, as git config must stay as is.

Note: git's `user.name` is `bmitigoa` (not "Baraka Mitigoa"). Commits carry that name. To change it later: `git config user.name "Baraka Mitigoa"`.

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9. Git log

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f31589c chore: scaffold Drupal 11 project with DDEV and Drush
91b1dc5 config: initial Drupal standard profile configuration export
41144d5 feat: add contrib modules (admin toolbar, pathauto, token, geofield, leaflet, charts,
address, devel)
275610e chore: add Day 1 content model build script
96e38bd config: add sector/region taxonomies, project/location/partner/report content types and
pathauto patterns
```

Two further commits followed: `docs: day 1 report` (4cb7a3a) added this report and its PDF, and `chore: untrack local tooling files` (d93d491) stopped tracking `CLAUDE.md`. The history was later rewritten to remove `CLAUDE.md` entirely, which is why these hashes differ from any earlier ones.

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10. Next day (Day 2 â€” Roles, Workflow & Sample Data)

- Create roles: Field Officer, M&E Reviewer, Admin, plus one test user per role.

- Enable Content Moderation and Workflows with Draft â†’ In Review â†’ Published â†’ Archived on Reports.
- Set permissions: field officers create/edit their own reports and submit for review, reviewers approve or reject, admins manage everything.
- Seed realistic data: about 10 projects, 15 Ugandan locations with real coordinates, 5 partners and 30 reports.